UNIT - V

PUMPS (CENTRIFUGAL PUMPS)

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Working principle of a centrifugal pump; work done by impeller
- Heads, losses and efficiencies; minimum starting speed; priming
- Specific speed; limitation of suction lift; NPSH
- Performance and characteristic curves; cavitation effects
- Multistage centrifugal pumps; troubles and remedies

1. Introduction and Working Principle

1.1 Pumps: classification

- A **pump** converts mechanical energy (from a motor / engine) into **hydraulic energy** (head) of the liquid.

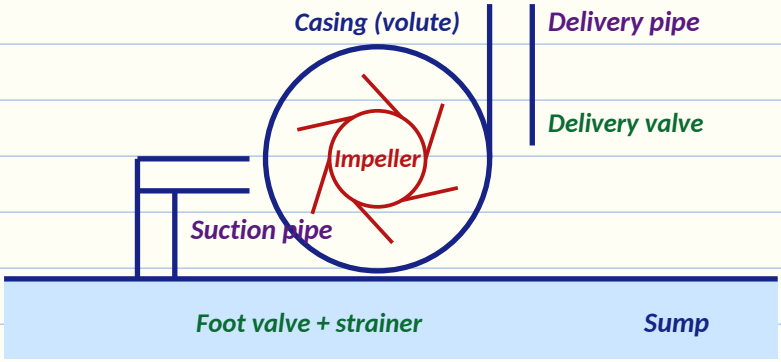
Type	Examples
Rotodynamic (kinetic)	Centrifugal (radial), mixed flow, axial flow pumps
Positive displacement	Reciprocating (piston, plunger), rotary (gear, vane, screw) pumps

- Centrifugal pumps are the most common: simple, compact, smooth flow, suitable for large discharge and moderate head.

1.2 Principle

A centrifugal pump works on the principle of **forced vortex flow**: when a mass of liquid is rotated by an impeller, a **centrifugal head** is created, so the liquid is thrown outward from the centre with increased pressure and velocity.

- Water is sucked into the **eye** of the impeller (low pressure at the centre), whirled outward by the vanes, collected in the casing (volute) where its kinetic energy is converted to pressure, and delivered through the delivery pipe.



Hs = suction head

Hd = delivery head

Hm = Hs + Hd + losses

1.3 Main parts

- **Impeller**: wheel with backward-curved vanes; types: open, semi-open, closed (shrouded); single or double suction.
- **Casing** : air-tight passage; types: **volute casing** (spiral, area increases gradually), **vortex casing** (circular chamber between impeller and volute), **casing with guide vanes (diffuser)** in turbine pumps.
- **Suction pipe** with **foot valve** (non-return) and **strainer** ; **delivery pipe** with **delivery valve** ; **shaft, bearings, stuffing box / gland** .

1.4 Heads

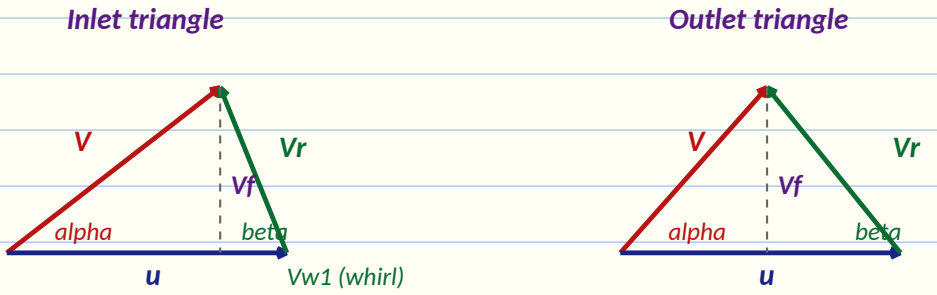
Term	Meaning
Suction head Hs	Vertical height of pump centre line above the sump water level
Delivery head Hd	Vertical height of the delivery outlet above the pump centre line
Static head Hst	Hs + Hd (total vertical lift)
Head loss	hfs (suction pipe) + hfd (delivery pipe) friction and minor losses
Manometric head Hm	Head actually developed by the pump: Hm = Hs + Hd + hfs + hfd + $Vd^2/2g$
Euler (theoretical) head He	Head given to the water by the impeller (before hydraulic losses)

$Hm = Hs + Hd + hfs + hfd + Vd^2 / 2g$; also $Hm = (\text{pressure head at delivery} - \text{pressure head at suction}) + (Vd^2 - Vs^2)/2g + \text{datum difference}$

2. Work Done by the Impeller

2.1 Velocity triangles

- **Inlet**: $u1 = \pi D1 N / 60$, Vf1 (radial), Vw1, Vr1, vane angle $\beta 1$ (with direction of u, backward). **Outlet**: $u2 = \pi D2 N / 60$, Vf2, Vw2, Vr2, outlet vane angle $\beta 2$.



Vw2 = u2 - Vf2 cot β2 (backward-curved vanes, β2 < 90°)

- For **radial entry** (no pre-whirl), α1 = 90°, Vw1 = 0. Flow rate **Q = π D1 B1 Vf1 = π D2 B2 Vf2** (B = width of impeller).

2.2 Euler's equation

Work done per second on water = ρ Q (Vw2 u2 - Vw1 u1)

Work done per unit weight (Euler head) He = (Vw2 u2 - Vw1 u1) / g ; for radial entry He = Vw2 u2 / g

- Backward vanes (β2 < 90°, 15° to 40°) give **stable operation and good efficiency**; radial vanes (β2 = 90°) give higher head but lower efficiency; forward vanes are not used.
- Pressure rise in the impeller (centrifugal head) = $(u2^2 - u1^2) / 2g + (Vr1^2 - Vr2^2) / 2g$.

2.3 Solved Problem 1

Impeller D2 = 0.4 m, N = 1450 rpm, B2 = 20 mm, β2 = 25°, Vf2 = 2.5 m/s, radial entry, ηman = 80%, ηo = 70%. Find Vw2, He, Hm, Q and shaft power.

- u2 = π x 0.4 x 1450 / 60 = **30.37 m/s** ; Vw2 = 30.37 - 2.5 x cot 25° = 30.37 - 5.36 = **25.0 m/s**.

He = Vw2 u2 / g = 25.0 x 30.37 / 9.81 = 77.4 m; Hm = 0.8 x 77.4 = *61.9 m*

Q = π D2 B2 Vf2 = π x 0.4 x 0.02 x 2.5 = 0.0628 m^3/s

Water power = ρ g Q Hm = 1000 x 9.81 x 0.0628 x 61.9 = 38.2 kW; Shaft power = 38.2 / 0.7 = *54.5 kW*

3. Losses and Efficiencies

3.1 Losses in a pump

Loss	Cause
Hydraulic losses	Friction in suction pipe, impeller and casing; shock (eddy) losses at inlet and outlet of vanes and at casing; diffusion losses
Mechanical losses	Friction in bearings, glands, stuffing box; disc friction (impeller rotating in fluid)

Leakage (volumetric) losses	Liquid flows back from delivery to suction through the gap between impeller and casing (wear ring clearance)

3.2 Efficiencies

Manometric efficiency $\eta_{man} = H_m / H_e = g\,H_m / (V_{w2}\,u_2)$

Volumetric efficiency $\eta_v = Q / (Q + q)$ (q = leakage) ; Mechanical efficiency η_m = (Power given to water by impeller) / (Shaft power)

Overall efficiency $\eta_o = (\rho\,g\,Q\,H_m)$ / (Shaft power) = $\eta_{man} \times \eta_v \times \eta_m$

- Typical overall efficiency of centrifugal pumps: 60 to 85% (large pumps up to 90%).
- Power input: $P = \rho\,g\,Q\,H_m / \eta_o$. Motor power is increased by 10 to 20% as a safety factor.

3.3 Solved Problem 2 (heads and power)

H_s = 4 m, H_d = 26 m, h_{fs} = 0.5 m, h_{fd} = 1.5 m, delivery velocity V_d = 2 m/s, Q = 0.05 m^3/s, η_o = 70%.

$H_m = 4 + 26 + 0.5 + 1.5 + 2^2 / 19.62 = 32.2$ m

$P = \rho\,g\,Q\,H_m / \eta_o = 1000 \times 9.81 \times 0.05 \times 32.2 / 0.70 = *22.6$ kW*

4. Minimum Starting Speed

4.1 Concept

- At start, the water in the impeller is **at rest and acts as a forced vortex**. The pump will start delivering only if the **pressure rise due to centrifugal action across the impeller $(u_2^2 - u_1^2)/2g$ is at least equal to the manometric head H_m**.

4.2 Derivation

Condition: $(u_2^2 - u_1^2) / 2g \geq H_m = \eta_{man}\,V_{w2}\,u_2 / g$ $\quad u_2^2 - u_1^2 = 2\,\eta_{man}\,V_{w2}\,u_2$

- Substituting $u = \pi\,D\,N / 60$: $(\pi\,N/60)^2\,(D_2^2 - D_1^2) = 2\,\eta_{man}\,V_{w2}\,(\pi\,D_2\,N / 60)$.

$N_{min} = 120\,\eta_{man}\,V_{w2}\,D_2 / [\,\pi\,(D_2^2 - D_1^2)\,]$ (rpm)

- Below N_{min} the pump cannot lift water to the delivery height (even if primed), so **the pump must run above this speed, with the delivery valve closed or partly open, at starting**.

Problem 3: D_2 = 0.4 m, D_1 = 0.2 m, V_{w2} = 15 m/s, η_{man} = 75%.

$N_{min} = 120 \times 0.75 \times 15 \times 0.4 / [\pi\,(0.16 - 0.04)] = 540 / 0.377 = *1432$ rpm*

5. Priming

- **Priming** = filling the suction pipe, casing and impeller **completely with the liquid** (expelling the air) before starting. Because air has very low density, a pump full of air produces negligible pressure rise and cannot lift water (**air binding**).
- **Methods**: pouring water through a funnel at the top of the casing (with foot valve), **self-priming pumps** , **ejector** , **vacuum pump** , jockey pump, or placing the pump **below** the water level (flooded suction, no priming needed).
- Starting sequence: prime, close the delivery valve (minimum power), start motor, open delivery valve slowly. At shut-off, close the delivery valve first.

6. Specific Speed

$N_s = N\sqrt{Q} / Hm^{(3/4)}$ (N in rpm, Q in m^3/s, Hm in m)

- **Specific speed of a pump = the speed of a geometrically similar pump that delivers 1 m^3/s against a head of 1 m.** It characterises the **shape** of the impeller.

Ns (approx., SI)	Pump type
10 to 30	Low-speed radial flow (narrow impeller, high head, low discharge)
30 to 80	Medium-speed radial (Francis-type)
80 to 160	High-speed / mixed flow
> 160	Axial flow (propeller) pumps: low head, large discharge

- Double-suction pump: use Q/2 in Ns. For multistage pumps, use the head per stage.

Problem 4: N = 1450 rpm, Q = 0.0628 m^3/s, Hm = 61.9 m. $N_s = 1450 \times \sqrt{0.0628} / 61.9^{0.75} = 1450 \times 0.2506 / 22.07 =$ **16.5** (low-speed radial impeller).

7. Limitation of Suction Lift and NPSH

7.1 Maximum suction lift

- Theoretically the suction lift cannot exceed **atmospheric head Ha = 10.33 m of water**. In practice it is limited to **about 6 to 7 m** because of vapour pressure Hv, friction losses hfs and velocity head in the suction pipe, and the pressure drop at the impeller entrance (cavitation).

Hs(max) = Ha - Hv - hfs - NPSHR (pump above the sump level)

7.2 Net Positive Suction Head

NPSH = total absolute head at the pump suction flange above the vapour pressure head, referred to the pump axis.

NPSH available (NPSHA) = Ha - Hv - Hs - hfs (Hs = suction lift; use - Hs for a lift and + Hs for flooded suction)

NPSH required (NPSHR) = a property of the pump (from the manufacturer's test curves; increases with Q and N)

- **Condition for cavitation-free operation: NPSHA > NPSHR** (with a margin of 0.5 to 1 m).

Thoma's cavitation number σ = NPSH / Hm (compare with σc)

- Remedies if NPSHA is low: lower the pump (reduce Hs), **increase pipe diameter** (reduce hfs), reduce flow temperature, reduce speed, use **inducer** or a pump with lower NPSHR, increase sump level.

Problem 5: Ha = 10.3 m, Hv = 0.24 m (water at 20°C), Hs = 3.5 m, hfs = 0.8 m, NPSHR = 4 m.

NPSHA = 10.3 - 0.24 - 3.5 - 0.8 = 5.76 m (> 4 m, safe margin 1.76 m)

Maximum suction lift = 10.3 - 0.24 - 0.8 - 4 = 5.26 m

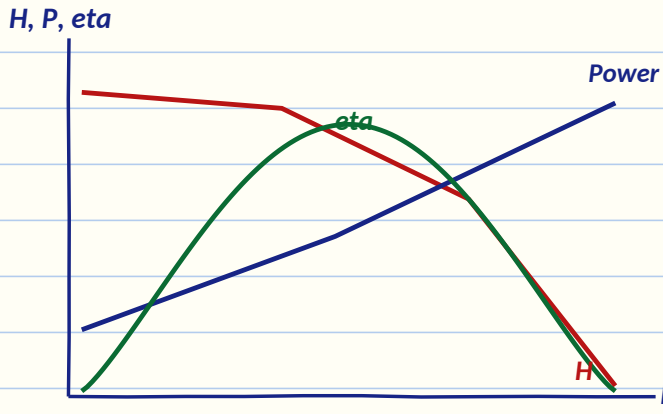
8. Cavitation in Pumps

- **Cavitation**: vapour bubbles form in the low-pressure region (impeller eye, back of vanes) when pressure falls to vapour pressure; they collapse as they move to high pressure.
- **Effects**: **noise and vibration** (crackling, gravel-like sound); **pitting and erosion** of impeller and casing; **fall in head, discharge and efficiency**; reduced pump life; possible shaft and bearing damage.
- **Prevention**: maintain NPSHA > NPSHR; low suction lift and short, straight, large suction pipe; avoid air leaks and sharp bends; use cavitation-resistant impeller materials (bronze, stainless steel); operate near the best efficiency point; keep liquid temperature low.

9. Performance and Characteristic Curves

9.1 Types of curves

Curve	Plotted	Constant
Main characteristic curves	Hm, P, η against Q at several speeds	Speed N varied
Operating characteristic curves	Hm, P, η against Q	Speed N constant
Constant efficiency (Muschel) curves	Contours of η on Hm-Q plane	η



Main characteristic curves:
H-Q falls as Q rises.
Power-Q rises with Q.
Efficiency peaks at the **best efficiency point (BEP).**

- **Hm-Q** : falls as Q rises (for backward vanes). **P-Q** : rises with Q (so start with delivery valve closed). **η-Q** : rises to a maximum (**best efficiency point, BEP**) then falls.
- The **shut-off head** is at Q = 0; the **runout** is at maximum flow.

9.2 Affinity laws (similarity)

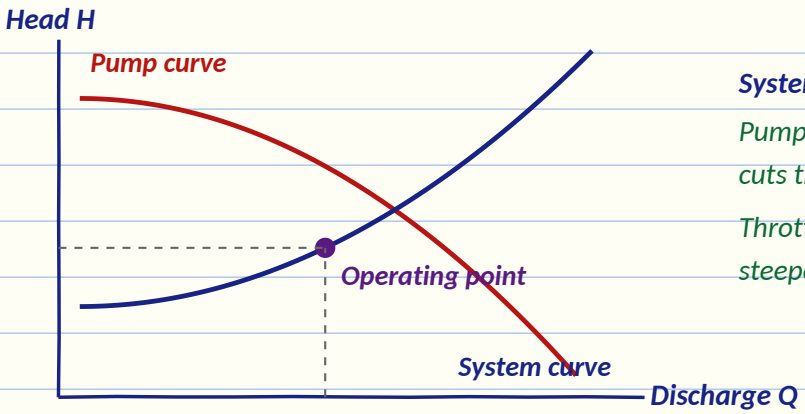
$Q \propto N D^3$; $Hm \propto N^2 D^2$; $P \propto N^3 D^5$ (same pump: $Q \propto N$, $H \propto N^2$, $P \propto N^3$)

Unit quantities: Nu = N / √Hm ; Qu = Q / √Hm ; Pu = P / Hm^(3/2)

Problem 6: At 1450 rpm a pump gives Q = 0.06 m^3/s, H = 40 m, P = 30 kW. Find values at 1750 rpm.

- Ratio r = 1750 / 1450 = 1.207. Q = 0.06 x 1.207 = **0.0724 m^3/s** ; H = 40 x 1.457 = **58.3 m** ; P = 30 x 1.758 = **52.7 kW** .

9.3 Operating point (pump and system)



System head = static head + K Q^2

Pump works where its H-Q curve cuts the system curve.

Throttling the delivery valve steepens the system curve.

- **System curve**: $H_{sys} = Hst + K Q^2$ (static head plus friction). The **operating point** is the intersection of pump and system curves.

Problem 7: Pump curve $H = 50 - 1500 Q^2$; system $H = 20 + 2000 Q^2$. Equate: $3500 Q^2 = 30$, so $Q^2 = 0.00857$, **$Q = 0.0926\ m^3/s$** *and* **$H = 37.1\ m$**.

10. Multistage Centrifugal Pumps

10.1 Need

- A single impeller gives limited head (about 80 to 100 m practical). For **high heads** (boiler feed, mines, high-rise buildings), several impellers are used in a single casing.

10.2 Pumps in series (multistage for head)



- Impellers are mounted on the **same shaft**, the **discharge of one stage enters the next**; **same Q, head adds**: $H = n \times H_{stage}$. **Guide vanes (diffusers)** between stages convert velocity to pressure and lead water to the next impeller eye.

10.3 Pumps in parallel (for discharge)

- Two or more impellers (or pumps) deliver into a **common delivery pipe**: **same head, discharge adds**: $Q = Q1 + Q2 + \ldots$ Used for large discharge at moderate head (low lift pumps); double-suction impeller is a parallel arrangement.

Arrangement	Head	Discharge	Use
Series (multistage)	Adds (n x H)	Same	High head

Parallel	Same	Adds (n x Q)	High discharge, standby

Problem 8: H = 120 m, Q = 0.05 m^3/s, N = 1450 rpm. Single stage: Ns = 1450 x √0.05 / 120^0.75 = 1450 x 0.2236 / 36.26 = 8.9 (impractically low). With 4 stages (30 m each): Ns = 1450 x 0.2236 / 30^0.75 = 1450 x 0.2236 / 12.82 = **25.3**, a good efficient radial impeller.

11. Troubles and Remedies

Trouble	Probable cause	Remedy
Pump does not deliver water	Not primed, air leak in suction, foot valve stuck, speed too low, suction lift too high	Re-prime, seal leaks, repair foot valve, check speed and rotation, reduce lift
Low discharge / head	Wrong direction of rotation, worn impeller or wear rings, partly clogged strainer, air entering gland	Correct rotation, replace impeller / rings, clean strainer, repack gland
Loses water after starting	Air leak, low NPSH, water level in sump falls	Fix leaks, raise sump level, increase pipe size
Motor overload / excess power	Speed too high, delivery valve fully open at low head, mechanical binding	Throttle delivery valve, check speed, realign shaft
Vibration and noise	Cavitation, misalignment, unbalanced impeller, loose foundation, damaged bearing	Raise NPSHA, align shaft, balance impeller, tighten bolts, replace bearing
Overheating of bearings / pump	Poor lubrication, running with valve closed, misalignment	Lubricate, avoid shut-off running, align
Leakage at gland	Worn packing or sleeve	Replace packing, tighten gland evenly
Pitting of impeller	Cavitation, corrosion	Improve suction conditions, use better material

12. More Concepts and Solved Problems

12.1 Centrifugal vs reciprocating pump

Basis	Centrifugal pump	Reciprocating pump
Discharge	Large, continuous and steady	Small, pulsating (needs air vessel)
Head	Low to medium (multistage for high)	Very high heads possible
Viscous / dirty liquid	Suitable for clean, moderately dirty	Not suitable for dirty liquids
Efficiency curve	Falls when head rises	Nearly constant (positive displacement)
Cost, space, maintenance	Compact, low cost, simple	Bulky, costly, more wear parts
Priming	Required (except self-priming)	Self-priming

12.2 Casing types

Casing	Features
Volute	Spiral passage with area increasing towards the delivery; velocity head converted to pressure; simple, common
Vortex	Circular chamber (vortex) between impeller and volute; improves efficiency slightly; larger size
Diffuser (turbine pump)	Fixed guide vanes surround the impeller; smoother conversion; used in multistage and high-head pumps

12.3 Mixed flow and axial flow pumps

- **Mixed flow pump**: liquid leaves the impeller partly radially and partly axially (Ns 80 to 160): medium head and large discharge. **Axial flow (propeller) pump**: flow parallel to the shaft (Ns > 160): large discharge, low head (drainage, irrigation, cooling water); has a rising power curve at low flow and a steep head curve.

12.4 Slip and slip factor

- The actual whirl velocity at outlet is **less than the theoretical** V_{w2} because of relative circulation of liquid in the impeller passages (**slip**). Slip factor $\sigma_s = V_{w2}' / V_{w2} < 1$; **Stodola**: $\sigma_s = 1 - 0.63\pi / Z$ (Z = number of vanes). For $Z = 8$: $\sigma_s = 0.75$.

12.5 Effect of vane outlet angle

Vane type	β_2	Head	Remarks

Backward curved	< 90°	Lower Euler head	Stable, high efficiency, power curve self-limiting
Radial	= 90°	Moderate	Used in small pumps and slurry service
Forward curved	> 90°	Highest Euler head	Poor efficiency, unstable, large kinetic energy loss

Problem 9: Impeller design

N = 1450 rpm, Hm = 40 m, Q = 0.1 m^3/s, ηman = 80%, speed coefficient $u2 = \sqrt{(2\,g\,Hm)}$, $Vf2 = 0.12\sqrt{(2\,g\,Hm)}$. Find D2, Vw2, β2 and B2.

- $\sqrt{(2\,g\,Hm)} = 28.0$ m/s, so u2 = 28.0 m/s; $D2 = 60\,u2 / \pi N = 60 \times 28.0 / (\pi \times 1450) =$ **0.369 m**.
- $He = Hm / \eta man = 50$ m; $Vw2 = g\,He / u2 = 9.81 \times 50 / 28.0 =$ **17.5 m/s**; Vf2 = 3.36 m/s.

$\tan \beta 2 = Vf2 / (u2 - Vw2) = 3.36 / 10.5 = 0.32 \quad \beta 2 = 17.7°$

$B2 = Q / (\pi\, D2\, Vf2) = 0.1 / (\pi \times 0.369 \times 3.36) = 0.0257$ m ≈ 26 mm

Problem 10: Pumps in parallel and series

Each pump: $H = 50 - 1500\,Q^2$; system: $H = 20 + 2000\,Q^2$. Find the operating point for (a) one pump, (b) two in parallel, (c) two in series.

- (a) Single: Q = 0.0926 m^3/s, H = 37.1 m.

(b) Parallel: $H = 50 - 1500\,(Q/2)^2 = 50 - 375\,Q^2$; equate with system: $30 = 2375\,Q^2$, Q = 0.112 m^3/s, H = 45.3 m

(c) Series: $H = 100 - 3000\,Q^2$; $80 = 5000\,Q^2$, Q = 0.1265 m^3/s, H = 52 m

- **Note**: discharge is not doubled in parallel because friction head increases with Q^2; series raises head, but flow rises only moderately.

Problem 11: Power and energy

Lift 100 m^3/h to a tank, Hm = 25 m, ηo = 65%.

- Q = 100/3600 = 0.02778 m^3/s; $P = 1000 \times 9.81 \times 0.02778 \times 25 / 0.65 =$ **10.5 kW**; energy for 8 h = 84 kWh.

Problem 12: Specific speed of double-suction pump

N = 980 rpm, Q = 0.4 m^3/s, Hm = 25 m, double suction (each side Q/2 = 0.2 m^3/s).

Ns = 980 x √0.2 / 25^0.75 = 980 x 0.447 / 11.18 = *39.2* (medium-speed radial pump)

Problem 13: Centrifugal pressure rise

Data of Problem 1 with D1 = 0.2 m: u1 = 15.2 m/s, u2 = 30.4 m/s.

Centrifugal head = $(u2^2 - u1^2) / 2g = (922 - 231) / 19.62 = 35.2$ m (compare with Hm; pump starts only if this is more than Hm, otherwise needs a higher speed, as in Problem 3).

13. Quick Revision and Important Questions

Euler: He = Vw2u2/g; Vw2 = u2 - Vf2 cotβ2. Hm = Hs + Hd + losses + $Vd^2/2g$. ηo = ηman ηv ηm. Nmin = 120ηman Vw2 D2/[π($D2^2$ - $D1^2$)]. Ns = N√Q/Hm^(3/4). NPSHA = Ha - Hv - Hs - hfs > NPSHR. Affinity: Q N, H N^2, P N^3.

- Explain the working of a centrifugal pump with a neat sketch. State its main parts. (10 M)
- Derive the expression for the work done by the impeller (Euler's head equation) with velocity triangles. (10 M)
- Define manometric head, and manometric, volumetric, mechanical and overall efficiencies. (5 M)
- Derive the expression for the minimum starting speed of a centrifugal pump. (5 M)
- What is priming? What is NPSH? Explain the limitation of suction lift. (5 M)
- Define specific speed. Explain performance (characteristic) curves of a centrifugal pump. (10 M)
- Write a note on cavitation in pumps and multistage pumps. List common troubles and remedies. (10 M)